

Distribution Theory

Functional Analysis

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Remark

We consider the measure space $(\mathbb{R}^n, \mathcal{L}(\mathbb{R}^n), \lambda^n)$ where $\mathcal{L}(\mathbb{R}^n)$ is the Lebesgue σ -algebra and λ^n is the Lebesgue measure on $\mathbb{R}^n$. Every integral of a function $f : \mathbb{R}^n \rightarrow \mathbb{C}$ is assumed to be taken over $\mathbb{R}^n$ and with respect to λ^n , i.e.

$$\int f \equiv \int_{\mathbb{R}^n} d\lambda^n f$$

unless otherwise specified.

Example 1.1

Consider the **Sobolev's mollifier** $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ defined as

$$\varphi(x) = \begin{cases} c \exp\left(-\frac{1}{1-x^2}\right) & |x| < 1, \\ 0 & |x| \geq 1, \end{cases}$$

where c is chosen in such way that $\int \varphi = 1$, and define $\varphi_{x_0,n}$ as

$$\varphi_{x_0,n} = n\varphi(n(x - x_0)).$$

Then for any f continuous we have

$$f(x_0) = \lim \int f \varphi_{x_0,n}.$$

Definition

The **support** of a function $f : \mathbb{R}^n \rightarrow \mathbb{C}$ is the closure of the region of $\mathbb{R}^n$ where $f(\mathbf{x}) \neq 0$. Following Heine and Borel, we will say that a complex-valued function f on $\mathbb{R}^n$ has **compact support** if the support is a closed and bounded set; that is, if there exists $R > 0$ such that $f(\mathbf{x}) = 0$ for $|\mathbf{x}| \geq R$.

Definition

We set $\mathcal{C}^m(\mathbb{R}^n)$ to be the space of all C^m functions on $\mathbb{R}$ with compact support, called the space of **test functions of order m** . The space $\mathcal{C}^\infty$, called the space of **test functions** is defined as

$$\mathcal{C}^\infty = \bigcap_{m \in \mathbb{N}} \mathcal{C}^m(\mathbb{R}^n).$$

Since we will be mostly working on $\mathcal{C}^\infty$, we will simply write $\mathcal{C}^\infty = \mathcal{C}$.

Test Functions

Distributions

Regular
DistributionsOperations on
DistributionsDirac's Delta
DistributionDifferentiation of
DistributionsChange of Variable
in Dirac's Delta

Fourier Transforms

Schwartz Space

Definition

The **uniform** (or **supremum**) **norm** is defined as

$$\|f\|_{\infty} = \sup_{\mathbf{x} \in \mathbb{R}^n} \{|f(\mathbf{x})|\}.$$

Definition

Let α be a multi-index. A sequence (φ_n) of test functions is said to **converge to order m to φ in $\mathcal{C}$** , written $\varphi_n \xrightarrow[m]{\mathcal{C}} \varphi$ if there is a compact subset $K \subset \mathbb{R}^n$ such that

$$\text{supp}(\varphi) \cup \bigcup_{n \in \mathbb{N}} \text{supp}(\varphi_n) \subset K$$

and if $D^\alpha \varphi_n \Rightarrow D^\alpha \varphi$ on K (both with respect to the uniform norm) for all $|\alpha| \leq m$, in symbols, if for all $n \in \mathbb{N}$ there exists $r > 0$ such that

$$\varphi_n(\mathbf{x}) = 0 \quad \text{and} \quad \varphi(\mathbf{x}) = 0 \quad \text{for} \quad |\mathbf{x}| \geq r,$$

and if for all multi-indices α with $|\alpha| \leq m$

$$\lim ||D^\alpha \varphi_n(\mathbf{x}) - D^\alpha \varphi(\mathbf{x})||_\infty = 0.$$

If (φ_n) converges to all orders to φ in $\mathcal{C}$, then we simply say that (φ_n) **converges to φ in $\mathcal{C}$** . In this case, we write $\varphi = \mathcal{C}\text{-}\lim \varphi_n$

Definition

An element $T \in \mathcal{C}^*$ of the dual space of $\mathcal{C}$ is said to be **sequentially continuous to order m** if $T(\varphi_n) \rightarrow T(\varphi)$ whenever $\varphi_n \xrightarrow[\mathcal{C}]{m} \varphi$ and called a **distribution of order m** , similarly, it is said to be simply **sequentially continuous** if it is sequentially continuous to all orders. If $T \in \mathcal{C}^*$ is sequentially continuous, then T is called a **distribution**. In particular, the zero functional on $\mathcal{C}^*$, denoted simply by 0, is continuous and called the **zero distribution**. The space of all elements of the dual space $\mathcal{C}^*$ of $\mathcal{C}$ that are also continuous is denoted by $\mathcal{C}'$ and called the **space of distributions**.

Proposition 1.2

A distribution of order m is automatically a distribution of order m' for any $m' \geq m$ since convergence of order m' implies convergence of order m for $m' \geq m$.

Theorem 1.3

T is a distribution if and only if for every compact $K \subseteq \mathbb{R}^n$ there exists $m \in \mathbb{N}_0, c > 0$ such that if $\text{supp}(\varphi) \subseteq K$, then

$$|T(\varphi)| \leq c \sum_{|\alpha| \leq m} \|D^\alpha \varphi\|_\infty$$

for all test functions φ .

Definition

Two distributions $T, M \in \mathcal{C}'$ are said to be equal if $T(\varphi) = M(\varphi)$ for all $\varphi \in \mathcal{C}$.

Remark

A measurable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is said to be **locally integrable** if it is Lebesgue integrable on every compact subset $K \subset \mathbb{R}^n$. The space of all locally integrable functions on $U \subseteq \mathbb{R}^n$ is denoted by $\mathcal{L}_{loc}^1(U)$, when $U = \mathbb{R}^n$ we will simply write $\mathcal{L}_{loc}^1$.

Theorem 1.4

Every $f \in \mathcal{L}_{loc}^1$ defines a distribution of order 0 (and thus of every order) T_f defined by

$$T_f(\varphi) \equiv \langle f | \varphi \rangle = \int f \varphi, \quad \varphi \in \mathcal{C}.$$

We call f the **density** of T_f .

Definition

*Distributions T having a density $T = T_f$ are called **regular distributions** while those not corresponding to any locally integrable function are called **singular**.*

Theorem 1.5 (The Fundamental Theorem of Calculus of Variations)

A regular distribution T_f is equal to 0 if and only if $f = 0$ a.e.

[Test Functions](#)[Distributions](#)[Regular
Distributions](#)[Operations on
Distributions](#)[Dirac's Delta
Distribution](#)[Differentiation of
Distributions](#)[Change of Variable
in Dirac's Delta](#)[Fourier Transforms](#)[Schwartz Space](#)

Remark

If T and S are distributions of order m on $\mathbb{R}^n$, then clearly $T + S$ and aT are distributions of this order for all $a \in \mathbb{R}$, and hence $\mathcal{C}'$ is a linear space.

Remark

The product ST of two distributions is not a distribution, for $TS(\varphi) = T(\varphi)S(\varphi)$ is not linear in φ . However, if α is at least class $C^m(\mathbb{R}^n)$ and T is a distribution of order m , then αT can be defined as a distribution of order m , by setting

$$(\alpha T)(\varphi) = T(\alpha\varphi).$$

If T is regular with density f , then $\alpha T_f = T_{\alpha f}$. In this case α need only be a locally integrable function.

Example 1.6

Define the map $\delta_a : \mathcal{C}(\mathbb{R}) \rightarrow \mathbb{R}$ by

$$\delta_a(\varphi) = \varphi(a).$$

Show that δ_a is a singular distribution. In particular, we write δ for δ_0 . This is called **Dirac's delta distribution**. The **n -dimensional delta distribution** $\delta_{\mathbf{a}}^n : \mathcal{C} \rightarrow \mathbb{R}$ may be similarly defined by

$$\delta_{\mathbf{a}}^n(\varphi) = \varphi(\mathbf{a}).$$

Remark

Dirac's delta distribution δ_a is singular. Nevertheless, physicists often maintain the density notation and write

$$\delta(\varphi) = \int \varphi \delta = \varphi(0).$$

In writing such an equation, we can formally understand this as

$$\delta(\varphi) = \int d\delta \varphi$$

where δ is the Dirac measure on $\mathbb{R}^n$ concentrated at $x = 0$. It is also common to write $\delta_a(x) = \delta(x - a)$, or in the n -th dimensional case, $\delta^n(x - a) = \delta(x_1 - a_1)\delta(x_2 - a_2)\cdots\delta(x_n - a_n)$.

Definition

A sequence (δ_n) in $\mathcal{L}_{loc}^1$ is called a **delta sequence** if

- i) $\delta_n \geq 0$ for all $n \in \mathbb{N}$;
- ii) $\int \delta_n = 1$ for all $n \in \mathbb{N}$;
- iii) $\delta_n(x) \Rightarrow 0$ in $\|x\| \geq r$ for all $n \in \mathbb{N}$ and any $r > 0$.

Theorem 1.7

If (δ_n) is a delta sequence, then $\lim \delta_n = \delta$ in the sense that

$$\lim \int \varphi \delta_n = \varphi(0)$$

for any test function φ .

Example 1.8

Consider the sequence

$$\delta_n(x) = \frac{n}{\sqrt{\pi}} e^{-n^2 x^2}$$

of Gaussian functions. Show that (δ_n) is a delta sequence.

Definition

The **derivative of a distribution** T of order $m \geq 0$ is the distribution T' of order $m + 1$ given by

$$T'(\varphi) = T(-\varphi') = -T(\varphi').$$

Test Functions

Distributions

Regular
Distributions

Operations on
Distributions

Dirac's Delta
Distribution

Differentiation of
Distributions

Change of Variable
in Dirac's Delta

Fourier Transforms

Schwartz Space

Theorem 1.9

Every distribution is infinitely differentiable

Theorem 1.10

Set θ to be the **Heaviside step function**

$$\theta(x) = \begin{cases} 1 & x \geq 0, \\ 0 & x < 0. \end{cases}$$

Then $T'_\theta = \delta$.

Theorem 1.11

For any test function φ , we have

$$\delta^{(m)}(\varphi) = (-1)^m \varphi^{(m)}(0).$$

[Test Functions](#)[Distributions](#)[Regular
Distributions](#)[Operations on
Distributions](#)[Dirac's Delta
Distribution](#)[Differentiation of
Distributions](#)[Change of Variable
in Dirac's Delta](#)[Fourier Transforms](#)[Schwartz Space](#)

Theorem 1.12 (Leibnitz Rule)

For any differentiable function α and distribution T we have

$$(\alpha T)' = \alpha T' + \alpha' T.$$

Remark

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous monotone increasing function such that $f(\pm\infty) = \pm\infty$, and we adopt Dirac's notation, then assuming integrals can be manipulated by the standard rules for change of variable,

$$\int \varphi(\delta \circ f) = \int \frac{\varphi \circ f^{-1}}{f' \circ f^{-1}} \delta = \frac{\varphi(a)}{f'(a)}$$

where $f(a) = 0$. If f is monotone decreasing, the integration is inverted, resulting in a sign change. The general formula for a monotone function f of either direction, having a unique zero at $x = a$, is

$$\int \varphi(\delta \circ f) = \frac{\varphi(a)}{|f'(a)|}.$$

Symbolically, we may write

$$\delta \circ f = \frac{1}{|f'(a)|} \delta_a.$$

Essentially this equation can be taken as the definition of the distribution $\delta \circ f$. Setting $f(x) = -x$, it follows that δ is “even” in the sense that $\delta(x) = \delta(-x)$. If f has zeroes

Definition

For any function φ its **Fourier transform** is the function $\mathcal{F}[\varphi]$ defined by

$$\mathcal{F}[\varphi](y) = \frac{1}{\sqrt{2\pi}} \int dx \, \varphi e^{-ixy}.$$

The **inverse Fourier transform** is defined by

$$\mathcal{F}^{-1}[\varphi](y) = \frac{1}{\sqrt{2\pi}} \int dx \, \varphi e^{ixy}.$$

Theorem 1.13 (Fourier's Integral Theorem)

Let φ be a function such that $|\varphi|$ is Lebesgue integrable and is of bounded variation, then

$$\mathcal{F}^{-1}[\mathcal{F}[\varphi]] = \varphi.$$

Remark

The Fourier transform of δ (using the density notation) should be

$$\mathcal{F}[\delta(x)](y) = \frac{1}{\sqrt{2\pi}} \int dx \, \delta(x) e^{-ixy} = \frac{1}{\sqrt{2\pi}},$$

so that we expect

$$\delta(x) = \mathcal{F}^{-1} \left[\frac{1}{\sqrt{2\pi}} \right] = \frac{1}{2\pi} \int dy \, e^{ixy}.$$

Definition

The **Fourier transform of a distribution** T is defined to be the distribution $\mathcal{F}[T]$ given by

$$\mathcal{F}[T(\varphi)] = T(\mathcal{F}[\varphi])$$

for all test functions φ .

Theorem 1.14

For regular distributions we have (due to Fubini's theorem)

$$\mathcal{F}[T_f(\varphi)] = T_{\mathcal{F}[f]}(\varphi).$$

Moreover, if the inverse Fourier transform is defined on distributions by

$$\mathcal{F}^{-1}[T(\varphi)] = T(\mathcal{F}^{-1}[\varphi]),$$

then

$$\mathcal{F}^{-1}[\mathcal{F}[T]] = T.$$

Definition

We define the **Schwartz space** (or the space of **rapidly decreasing functions**) as

$$\mathcal{S}(\mathbb{R}) = \left\{ \varphi : \sup_{x \in \mathbb{R}} |x^m D^p \varphi(x)| < \infty \text{ for all positive integers } m, p \right\}.$$

Convergence in $\mathcal{S}(\mathbb{R})$ is defined by $\varphi_n \xrightarrow{\mathcal{S}} \varphi$ if and only if

$$\limsup_{x \in \mathbb{R}} |x^m (D^p \varphi_n(x) - D^p \varphi(x))| = 0$$

for all positive integers m, p . The space of continuous linear functionals on $\mathcal{S}(\mathbb{R})$ is denoted by $\mathcal{S}'(\mathbb{R})$, and they are called **tempered distributions**

Test Functions

Distributions

Regular
Distributions

Operations on
Distributions

Dirac's Delta
Distribution

Differentiation of
Distributions

Change of Variable
in Dirac's Delta

Fourier Transforms

Schwartz Space

Corollary

$$\mathcal{C}(\mathbb{R}) \subset \mathcal{S}(\mathbb{R}).$$

Remark

The Fourier transform of the delta distribution is defined by

$$\mathcal{F}[\delta_a(\varphi)] = \delta_a(\mathcal{F}[\varphi]) = T_{(2\pi)^{-1/2}e^{-iax}}(\varphi).$$

Similarly

$$\mathcal{F}^{-1}[T_{e^{-iax}}] = \sqrt{2\pi}\delta_a.$$